Principles of Testing:

1. Testing shows the presence of defects, not their absence.

* Testing is only shows the ~~Error~~, ~~and~~ defects are present in the software. But it cannot prove. that there are no defects.

2. Exhaustive testing is Impossible.

* It's Impossible to test all possible inputs and conditions.

* Therefore only Important test cases are Tested.

3. Early Testing.

* Testing should start the early as possible in the software development process to find defects early.

4. Defect Clustering.

Most defects are Usually found in a small Number of Modules.

5. Pesticide paradox :

If the same Test Cases are Repeated many times, new defects will not be found. so New test cases should be created.

6. Testing is context dependent :

Testing depends on the context and type of software Being tested.

for Example; the Testing Required the ecommerce site will be differ from the testing for a simple mobile game.

7. Absence of errors fallacy.

Even if no defects are found, the software must meet the users requirements.

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QA (Quality Assurance):

Quality Assurance is the process of ensuring that the software development process follows proper standards and methods to produce a high quality product.

QC (Quality Control)

Quality control is the process of testing the final software product to identify defects and ensures the product meets quality requirements.

Differ from QA and QC

	QA	QC
Approach	QA follow a process oriented approaches to ensures proper development procedures.	QC follow a product oriented approach to detect defects in the final product.

Focus	focusing on Improving the process development	focusing on the Quality of the final product.
Goal	Goal is to prevent defects	Goal is Identify a defects.
Timeline	QA activities are starts from beginning of the software development process.	QE activities are happen after the product is developped.
Responsiblity	The Entire team are Responsible for the QA.	Testing Team is Responsible in QC.
Activities.	Activities Included process review, Audits and documentation Checks.	Activities Included testing, defect Identification, and bug Reporting.